

Quantum-Enhanced Agrivoltaic Digital Twin: Spectral Co-Design, Multifunctional Net Energy Benefit, and Quantum-Secured Telemetry

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.32 yr, identifying conditions under which the multifunctional net benefit stays positive for the modeled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics

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1. Introduction

Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1, 2]. Land-use optimization and mobile-PV foundations for this field were established in *Applied Energy* [3, 4]. Recent syntheses systematize overhead, vertical, and tracking typologies with land-equivalent-ratio and levelized-cost metrics [5]. Across these studies, three gaps recur and remain unresolved. The optical architecture is *static*: semi-transparent PV claddings act as fixed spectral filters that attenuate irradiance uniformly, trading electrical output for crop biomass without optimization of the spectral light-sharing window [6, 7]. Dynamic, physics-guided spectral management has not been transposed into an operating agrivoltaic stack. The cost of stasis is quantifiable: in the mode-volume scan reported below, the forward transfer yield varies by a factor of 1.6 and the SERS enhancement by a factor of 48 across the accessible range (Table 2), so a statically chosen cladding fixes the energy–diagnostics balance at deployment with no ability to re-trade it. The scan probes the strong-coupling penalty only: all seven volumes carry the NPoM channel, and the filtered-excitation gains quantified next apply to the NPoM-free fraction at every point of the scan. Techno-economic and lifecycle assessments remain decoupled from the underlying device physics: electricity, crop, water, and carbon accounts are evaluated subsystem by subsystem, so multifunctional trade-offs are described rather than co-optimized [8, 9]. The cost of this decoupling is concrete: accounting that ignores the local transport physics would credit the canopy with the passive baseline of 0.980 instead of the physically consistent 0.971 — and would never discover that the 91.8 % local penalty costs only 0.9 points globally, a stabilizing margin that makes NPoM integration viable. The PV layer is likewise treated purely as an energy harvester: in-situ crop-stress and contaminant sensing is bolted on as separate hardware rather than co-designed with the optical stack, despite the maturity of field-deployable SERS analysis [10], of quantum-enabled agricultural workflows [11], and of quantum-sensing readiness for precision agriculture [12]. Co-design changes the economics of the sensing function: reusing the NPoM optical infrastructure delivers SERS diagnostics at a canopy-area cost of 1 % and a canopy-yield cost of 0.9 % relative to the passive baseline (0.980 to 0.971), instead of separate sensing hardware with its own capital and footprint.

The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [13]. This spectral specificity

lets the PV layer transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a problem of spectral co-design.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as the model system for spectral light-sharing [14]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [15] and room temperature [16]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic transitions with underdamped protein vibrations — governs the flow of excitation energy [17, 18]. Non-Markovian frameworks such as the hierarchical equations of motion [19] and the process tensor hierarchy of pure states (PT-HOPS) [20, 21] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare: the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic product [22]. Photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [23, 24].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [25]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. That companion paper, published in the *Journal of Physical Chemistry Letters*, reports the analysis of the filter, the excitation-transport rate, and the single-pass conversion efficiency [25]; the present work adds the 8-site trapping model, the monolithic OPV–NPoM hardware concept, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [26], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [27]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror

(NPoM) plasmonic cavity [28] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and flux-dependent optical limiting — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman-Monteith model [29] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [30] for secure IoT telemetry. This multi-scale perspective [31] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modeling framework for quantum-guided agrivoltaics (Fig. 5).

The central question of this work is falsifiable: *does an agrivoltaic stack whose spectral architecture is co-designed by exact non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (power + crop + water + sensing) under field constraints, and which deployment barriers are physical rather than economic?* Three subordinate questions structure the Results. (Q1) Device asks what the NPoM co-design costs and where the cost lands, quantified by the mode-volume scan (Table 2) and located by the canopy-integration analysis. (Q2) Physics asks which microscopic states carry the trade-off: exact PT-HOPS dynamics localize the loss channels, and Floquet-Stark detuning with flux-dependent optical limiting tests whether they can be managed dynamically. (Q3) System asks whether the stack pays for itself, once coupled FAO-56 microclimate, cooperative techno-economics, and BB84-secured telemetry internalize every function into a single Net Ecological Benefit account. Three falsifiable claims anchor the answer. The first, (F1) Microscopic, is that dual-band spectral filtering at 750 nm and 820 nm prepares polaron-dressed states that extend exciton coherence by 50 % and raise the forward transfer yield by 25 % at 295 K. The second, (F2) Device, is that confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS diagnostics, despite the $> 90\%$ local transport penalty quantified across the mode-volume scan. The third, (F3) System, is that the FAO-56 smart shield cuts evapotranspiration by 28 % ($\pm 10\%$ microclimate uncertainty) and, combined with OPV generation and cooper-

116 active economics, yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and
 117 a 4.32 yr payback at 30 % subsidy for the modeled 500 m^2 cooperative case.
 118 We answer affirmatively but conditionally: the bounds (single-exciton treat-
 119 ment, site-independent baths, modeled fouling and matrix penalties, $\pm 10 \%$
 120 microclimate band) are stated explicitly in the Limitations section of the Dis-
 121 cussion. The headline outcome, pictured in [Figure 1](#), compresses this answer
 122 to one sentence: a 91.8 % local transport penalty confined to a 1 % sentinel
 123 area costs 0.9 points of canopy-scale trapping yield (0.980 to 0.971) while
 124 adding single-complex SERS diagnostics to the same footprint. To keep a
 125 stack of this breadth auditable, each layer — quantum dynamics, microcli-
 126 mate, economics, and security — is validated separately in the Supporting
 127 Information ([Section S2–Section S12.9](#)), and every headline number below is
 128 traceable to a single figure or table.

129 2. Results

130 2.1. 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1})**. The near-
 degenerate antenna pair (Sites 1–2) feeds the trap funnel (Sites 3–4) through the strongly
 coupled 4–5–6 backbone; Site 8 extends the antenna manifold with couplings an order of
 magnitude below J_{12} (strongest link $J_{18} = 21.0 \text{ cm}^{-1}$), so the 7-site transport references
 carry over to the 8-site network. These couplings set the transport timescale that the
 spectral filter exploits. Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	−87.7	30.0	−53.5	−70.7	81.1	39.7	12.0

131 We construct an 8-site model of the FMO complex, extending the stan-
 132 dard 7-site framework [\[32\]](#) by including the eighth bacteriochlorophyll, which
 133 extends the antenna manifold above Site 3 ($\varepsilon_8 = 200 \text{ cm}^{-1}$ relative to Site 3)
 134 with couplings ranging from $|J_{38}| = 0.6 \text{ cm}^{-1}$ to $J_{18} = 21.0 \text{ cm}^{-1}$ — an order
 135 of magnitude below the J_{12} backbone. The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} = 0.15 \text{ fs}^{-1}$ is the effective irreversible reaction-center trapping rate, corresponding to a capture time $\tau_{\text{trap}} \approx 6.7 \text{ fs}$. This capture time is short compared with the 1-ps propagation window, so near-complete capture (the 0.98 passive baseline) is dynamically accessible in [Eq. \(4\)](#) [33]. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. We verify that all 28 inter-site couplings are finite and that the non-Hermiticity is confined to the trapping diagonal terms ([Section S1](#)).

2.2. NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [28], where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to the FMO optical transitions and mediates the optical interface with the OPV layer.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [28]. At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1} to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [34].

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . [Figure 2a](#) shows the energy level diagram.

168 2.3. Non-Markovian quantum dynamics

169 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
 170 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
 171 terms, time step $\Delta t = 0.2$ fs, and $n = 100$ disorder realizations ($\sigma = 50$ cm $^{-1}$
 172 static disorder) [20, 21]. The bath is modeled as a Drude-Lorentz over-
 173 damped component ($\lambda_D = 35$ cm $^{-1}$, $\gamma_D = 50$ cm $^{-1}$) plus 12 underdamped
 174 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
 175 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
 176 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$ and
 177 $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is 3.3×10^{-5}
 178 (Section S2). Our companion study details and validates the computational
 179 feasibility, scaling, and memory-aware scheduling of these high-order simula-
 180 tions [25].

181 Figure 2b shows the exciton population dynamics of the sentinel NPoM
 182 configuration ($V = 1.2$ nm 3 , $n = 20$): the plasmon mode loads within 10
 183 fs and captures 82% of the excitation, and the reaction-center population
 184 saturates at the strongly suppressed value reported in Table 2. The forward
 185 transfer yield quantifies this capture:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

186 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

187 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$
 188 for broadband excitation — a relative enhancement of 25% (Table S4; par-
 189 enthetical uncertainties are 1 standard deviation over the $n = 100$ disorder
 190 ensemble). These filtered- and broadband-excitation reference trajectories
 191 (bare FMO, no NPoM channel) are inherited unchanged from our compan-
 192 ion study, which reports their full trajectory sets, computed on the seven-
 193 site network with an otherwise identical PT-HOPS/SBD protocol [25]; the
 194 eighth site added here couples an order of magnitude below the J_{12} back-
 195 bone (strongest link $J_{18} = 21.0$ cm $^{-1}$; weakest link into the trapping pair
 196 $J_{38} = 0.6$ cm $^{-1}$), so we carry the references over unchanged. The 8-site no-
 197 NPoM reference $\Phi_{\text{FT}} = 0.980$ is the assumed passive capture efficiency of
 198 the unimpeded network — a model input representing near-complete charge
 199 separation, not a trajectory integral: within the 1-ps window the off-cavity
 200 integral does not saturate, because the trapping sites retain steady-state pop-
 201 ulation (Section S11). These references characterize the 99% passive fraction

202 of the canopy and are therefore independent of, and complementary to, the
 203 NPoM mode-volume scan below, which probes the strong-coupling penalty
 204 on the sentinel fraction. The coherence lifetime τ_c , defined as the $1/e$ de-
 205 cay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [35], extends from
 206 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase that is
 207 causally linked to the yield gain. The extended coherence suppresses the
 208 pure-dephasing rate, lengthening the effective transport mean-free path and
 209 increasing the fraction of the excitation wavepacket that reaches the reaction
 210 center before decohering. The mechanism is identical to that established for
 211 the 7-site system [25]: the red passband is quasi-resonant with the trapping-
 212 site pair (sites 3/4 at $12\,210\text{ cm}^{-1}$ and $12\,320\text{ cm}^{-1}$, i.e. 819 nm and 812 nm;
 213 the 100 cm^{-1} passband is centred on site 3 and lies within 75 cm^{-1} of site 4),
 214 so the manifold members that bridge those sites to the bright exciton — the
 215 180 cm^{-1} Franck–Condon mode of site 4 and the 280 cm^{-1} mode of site 3
 216 — are the dressed set. Removing them leaves a residual reorganization en-
 217 ergy $\lambda_{\text{res}}^{\text{filt}} = 110\text{ cm}^{-1}$ instead of the full-bath $\lambda_{\text{res}} = 127.6\text{ cm}^{-1}$ (Section S3);
 218 because pure dephasing scales as $\lambda_{\text{res}}^{-2}$, this predicts a $\times 1.34$ coherence exten-
 219 sion, consistent with the computed $\times 1.50(18)$ ($280\text{ fs} \rightarrow 420\text{ fs}$) within the
 220 1σ lifetime uncertainty. The blue passband lies between the 720 cm^{-1} and
 221 $1\,050\text{ cm}^{-1}$ manifold members and acts as a transparent control window, con-
 222 sistent with the companion band scan in which moving the blue edge from
 223 730 nm to 770 nm left Φ_{FT} unchanged.

224 2.4. NPoM cavity volume dependence of exciton transport

225 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 226 active layer through the strong coherent interaction described above, yet the
 227 plasmonic mode also acts as an additional population decay channel compet-
 228 ing with reaction-center trapping. To quantify this trade-off, we perform a
 229 sensitivity scan of the NPoM mode volume V_{mode} , which controls the single-
 230 emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller mode
 231 volumes produce stronger plasmon–exciton coupling but also enhance the
 232 plasmon’s role as a population sink, diverting excitation away from the reac-
 233 tion center.

234 Table 2 summarizes the forward transfer yield across seven mode volumes
 235 spanning the experimentally accessible sub-nanometer range [28]. The yield
 236 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 237 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2\text{ nm}^3$, with a slight

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2 \text{ fs}$, $t_{\text{max}} = 1 \text{ ps}$. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [28]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	$\Delta \text{ vs. baseline (\%)}$	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

238 decrease at 1.4 nm^3 . The trend is logarithmic, with diminishing returns above
239 1.0 nm^3 : increasing the volume further recovers only 1 % of the baseline yield.

240 The physical origin of this suppression is the strong-coupling-induced hy-
241 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
242 a finite plasmonic character, which decays via the fast plasmon relaxation
243 pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
244 At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the
245 lower polariton branch reaches $\sim 15 \%$, diverting a corresponding fraction of
246 the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting
247 narrows, and the excitonic character of the bright states is partially restored.

248 Despite the monotonic recovery, all NPoM-on yields remain suppressed
249 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$).
250 Hence, in the strong collective-coupling regime accessible to NPoM cavities
251 ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally limits the
252 maximum achievable transport efficiency. The SERS enhancement factor
253 ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$, following the
254 Purcell enhancement of the local field at both the excitation and emission
255 wavelengths. Table 2 includes the calculated SERS enhancement for each
256 mode volume, revealing a four-order-of-magnitude range (33 to 1.6×10^3)

257 across the scanned volumes. The optimal volume represents a compromise:
 258 $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while retaining
 259 SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$). The tabulated EF_{SERS} values are a *relative*
 260 Purcell scaling normalized to 10^2 at $V_{\text{mode}} = 0.8 \text{ nm}^3$, not absolute single-
 261 molecule enhancements: absolute SERS in sub-nanometer NPoM gaps is
 262 measured at 10^6 and above [36], so single-complex detection of the 180 cm^{-1}
 263 Franck–Condon mode remains within the demonstrated capability of this ge-
 264 ometry [10]. This compromise only works because of the sentinel architecture
 265 — no volume on the scan simultaneously restores transport and maximizes
 266 diagnostics, which is precisely why the 1 % area split, not a better volume, is
 267 the design lever. By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity
 268 (1.6×10^3) at the cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off
 269 better suited to diagnostic-only applications where photochemical efficiency
 270 is secondary. At the reference volume, a dedicated $n = 100$ production run
 271 at $V_{\text{mode}} = 0.8 \text{ nm}^3$ confirms the same limiting behavior ($\Phi_{\text{FT}} = 0.0768$, final
 272 plasmon population 0.79; Section S11).

273 Reducing the temperature to 77 K (analyzed as a theoretical limit-case
 274 for thermal phonon suppression in Section S11.2) doubles the trapping yield
 275 to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K, the
 276 maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for $V_{\text{mode}} =$
 277 1.2 nm^3 , requiring the multi-scale sentinel fraction strategy to bypass plas-
 278 monic loading.

279 All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho =$
 280 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.
 281 The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations;
 282 the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} =$
 283 0.0799 , within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off
 284 run ($n = 20$) is shown as a limiting-case reference; quantitative comparison
 285 across configurations is limited by the differing trajectory counts.

286 2.5. Global canopy yield integration

287 Because the NPoM SERS diagnostic mode suppresses local excitonic
 288 transport through plasmon population loading, it need only cover a 1 % sen-
 289 tinel fraction α of the canopy area. The remaining 99 % operates in passive
 290 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 291 0.980 . The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

292 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 precisely);
 293 device-level excitation transfer is thus maintained near baseline even though
 294 transport is suppressed locally on the sentinel. $\Phi_{\text{FT}}^{\text{global}}$ is a device-level met-
 295 ric: crop productivity is governed by the photosynthetically active radiation
 296 transmitted through the module (the AVT/PAR budget), not by excitonic
 297 trapping in the FMO layer — the two are optically coupled only through the
 298 shared passband design, which is why no coherence-derived gain is booked
 299 in the crop revenue. Because Equation (5) is linear in α , the flatness of
 300 Figure 1b is an accounting identity rather than an independent dynamical
 301 result; what carries physical content are the two reference endpoints (0.080,
 302 measured on the sentinel run, and 0.980, the passive capture-efficiency cap)
 303 that enter it. The identity converts those endpoints into a design envelope:
 304 solving Equation (5) for α gives a sentinel fraction of up to 3.3 % before the
 305 global yield falls below 0.95, and up to 8.9 % before it falls below 0.90. The
 306 diagnostic area therefore has a factor ~ 3 of headroom over the 1 % assumed
 307 here — the co-design lever is not a better mode volume but how much area
 308 one is willing to spend on sensing.

309 2.6. Dynamic photo-protection via Floquet–Stark detuning and flux- 310 dependent optical limiting

311 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 312 changes dynamically, inducing a time-dependent Stark shift of the excitonic
 313 levels that detunes the OPV–FMO coupling. We model this as a Floquet
 314 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

315 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. Above thresh-
 316 old, the detuning shifts the OPV exciton energies out of resonance with the
 317 FMO absorption bands. With $V_0 = 0.05 \text{ eV} = 403 \text{ cm}^{-1}$ — $4.6\times$ the largest
 318 inter-site coupling $|J_{12}| = 87.7 \text{ cm}^{-1}$ — the single-channel transmission fac-
 319 tor of the resonant site-1 and site-6 pathways falls to $J^2/(J^2 + V_0^2) \approx 0.05$,
 320 i.e. these two antenna channels are effectively closed while the six alternative
 321 pathways through the network remain open (Figure 3a). To prevent rapid,
 322 noise-induced oscillations under fluctuating cloud cover, we implement a hys-
 323 teresis buffer of $\pm 50 \text{ W m}^{-2}$ (the detuning switch activates at 850 W m^{-2} and
 324 deactivates only when the flux drops below 750 W m^{-2}).

325 Concurrently, the NPoM cavity exhibits a flux-dependent optical-limiting
 326 response [37] that suppresses transmission at the 750 nm and 820 nm channels

under high flux — the intensity-dependent counterpart of the optomechanical modulation of the picocavity resonance (we use “optical limiting” rather than “transparency” deliberately: the engineered observable here is a *reduction* of passband transmission with flux). The transmission is modulated as:

$$T_{\text{OL}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the transmission drops to 44 % of baseline (Figure 3b), providing dynamic protection against exciton overload. At zero flux (night mode), both mechanisms return to zero without singularities (Section S4).

2.7. *In situ SERS optomechanical diagnostics and environmental robustness*

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39] — the integration gap addressed here. Multilayer graphene–metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [40]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode volumes.

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we assign three spectroscopic target signatures to the sentinel sub-system, plus a fourth, non-spectroscopic readout handled by a separate NV-diamond microfluidic assay. Oxidative stress is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. Pesticide contamination by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$ under controlled conditions), leveraging the plasmonic hot-spot amplification in the NPoM gap [10]. Heavy metal contamination of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots sealed within the NPoM sensor-probe polymer matrix ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$ under controlled conditions) [41]; unlike bare gold surfaces, the encapsulated probe is immune to dust and

360 biofouling. Cross-sensitivity between soil moisture and Pb^{2+} contamination
 361 is resolved by dual spectral interrogation: heavy metals modify the Stern–
 362 Volmer quenching intensity and fluorescence decay time, whereas soil mois-
 363 ture alters the local dielectric constant, producing spectral broadening or-
 364 thogonal to the metal-induced quenching signature. Pathogen detection uses
 365 Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microflu-
 366 idic “Point-of-Care” (Lab-on-a-Chip) desktop reader at the greenhouse hub
 367 (Section S12.7). These solid-state quantum sensors detect paramagnetic bac-
 368 terial metabolites from sap or irrigation water samples at pre-symptomatic
 369 stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural ma-
 370 trix will degrade in-field sensitivity, a loss mitigated by the graphene barrier
 371 and *DynamicCalibrator* corrections).

372 Figure 3c shows the simulated Raman spectrum for the three BChl *a*
 373 modes under active NPoM coupling. The relative intensities encode the in-
 374 stantaneous exciton population distribution, providing a direct optical probe
 375 of the energy transfer pathway.

376 Environmental robustness under realistic agricultural conditions (dust,
 377 bird droppings, microbial fouling) is modeled at three infrastructure levels.
 378 At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with
 379 days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

380 which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning
 381 interval (reached after 50 d); the 800 USD yr^{-1} cleaning line of the coopera-
 382 tive OPEX corresponds to seven or eight interventions per year at ~ 110 USD
 383 each, i.e. a schedule that never reaches the floor. At the sensor level, the
 384 graphene monolayer spacer of each NPoM cavity is functionalized with a
 385 zwitterionic polymer coating [42, 43], modeled in our simulations to pre-
 386 serve $\geq 90\%$ quantum coherence after 60 d continuous field immersion; it
 387 is the primary anti-fouling barrier, physically repelling microbial adhesion
 388 and protein adsorption. A thin-film micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of
 389 SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further at-
 390 tenuates the diurnal temperature excursion at the quantum probe, reducing
 391 the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$
 392 (Section S12.5). The *DynamicCalibrator* module — an exponential moving
 393 average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety
 394 net: it applies a multiplicative correction factor and raises a drift alarm when

the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

2.8. Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [29] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the array-level OPV shading factor $f_{\text{shade}} = 0.41$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient. All annual water, carbon, and cash figures quoted in this work are produced by the annual FAO-56 and LCA accounting chain described in the Methods; the orchestrator’s per-tick digital-twin water balance is an operational monitor on the same FAO-56 path, not an independent derivation. The shading factor is not a free parameter: it is the product of the module’s ground-coverage ratio and its complemented average visible transmittance, $f_{\text{shade}} = \text{GCR}(1 - \tau_{\text{AVT}}) = 0.55 \times 0.75 = 0.41$, so the canopy-level attenuation and the 15 %/25 % efficiency–transmittance module assumption (Section 5.1) describe the same stack.

Under standard conditions — clear-sky reference day with $R_n = 14.05 \text{ MJ m}^{-2} \text{ d}^{-1}$ (0.677 of the $5.76 \text{ kWh m}^{-2} \text{ d}^{-1} = 20.74 \text{ MJ m}^{-2} \text{ d}^{-1}$ daily horizontal irradiance after albedo and net longwave loss, consistent with a 600 W m^{-2} midday peak), 25°C , 60 % humidity, 2 m s^{-1} wind — the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.26 mm d^{-1} , i.e. $460 \text{ L m}^{-2} \text{ yr}^{-1}$ when annualized over 365 d (Figure 4a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [29].

2.9. Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at $8.2\text{ kg m}^{-2}\text{ yr}^{-1}$), reducing the entry-level capital expenditure to $161\,000\text{ USD}$ (322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the Monte-Carlo technoeconomic logic established for European agrivoltaic configurations [45]. A blended-finance subsidy of 30 % reduces the net CAPEX to $112\,700\text{ USD}$.

Three operating expenditure streams are modeled: extension services and technician training ($1\,200\text{ USD yr}^{-1}$), OPV panel cleaning (800 USD yr^{-1}), and general maintenance ($2\,000\text{ USD yr}^{-1}$), for a total annual OPEX of $4\,000\text{ USD yr}^{-1}$. The net annual cash flow of $26\,075\text{ USD yr}^{-1}$ (from $30\,075\text{ USD yr}^{-1}$ total revenue minus $4\,000\text{ USD}$ of OPEX) yields a simple payback period of 4.32 yr (or 6.17 yr without subsidy) and a 10-year discounted NPV of $+34\,630\text{ USD}$ at a 12 % risk-adjusted discount rate that we adopt as the Sub-Saharan infrastructure-finance assumption (Figure 4c). Holding other inputs fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback to 3.46–5.18 yr; a 50 % higher OPEX ($6\,000\text{ USD yr}^{-1}$) gives 4.49 yr, and 20 % lower revenue gives 5.61 yr. Beyond one-at-a-time variations, a seeded Monte Carlo (10^5 iterations, seed 42, the same engine as the in-pipeline uncertainty band) co-samples all inputs jointly: 10 % each on excitation yield, generation, lifecycle footprint, CAPEX, and revenue, 15 % on the water saving, and 10 % on the grid emission factor. It returns $\text{NEB} = 72.6$ with a 95 % interval of $54.3\text{--}92.2\text{ kg CO}_2\text{e m}^{-2}\text{ yr}^{-1}$, a payback mean of 4.38 yr (the deterministic point estimate is 4.32 yr; the ratio estimator is slightly convex) with 3.36–5.62 yr, and an NPV mean of $+34\,700\text{ USD}$ with a 95 % interval of $700\text{--}68\,600\text{ USD}$ — the 5th-percentile NPV stays positive, so the investment case survives the joint input scatter, not only the single-factor sensitivities (Section S5). A baseline CAPEX of 322 m^{-2} is a significant barrier in Sub-Saharan Africa relative to local income levels. Local construction materials (bamboo and timber framing instead of structural steel) and green chemical synthesis of gold nanoparticles from local plant extracts can nonetheless reduce the capital cost to 140 m^{-2} , lowering the unsubsidized payback period to 2.68 yr and making the modeled system potentially viable for local cooperatives under these cost assumptions. For benchmarking against the agrivoltaic technoeconomic literature [45, 9, 8], the 322 USD m^{-2} (2.15 USD W^{-1} of peak capacity at 15 % module efficiency) is dominated by conventional photovoltaic and balance-of-system content ($80\text{ OPV} + 30\text{ fiber} + 35\text{ installation} + 25\text{ electronics}$ of the 290 USD m^{-2}

subtotal); the NPoM sensing layer (120 USD m^{-2}) is the only quantum-specific line item, and its placeholder price is the assumption most in need of vendor quotation. A one-at-a-time stress test bounds that single placeholder: doubling the NPoM line item to 240 USD m^{-2} raises the module cost to 455 USD m^{-2} (410 subtotal plus the 11 % contingency) and lengthens the subsidized payback from 4.32 yr to 6.11 yr — still below the 6.17 yr unsubsidized payback of the baseline design, so a 100 % overrun on the only quantum-specific cost line does not reverse the investment case at 30 % subsidy. Two levelized metrics frame the same cash flow. First, levelized cost of electricity over the 10-year horizon: with the capital recovery factor $\text{CRF}(r = 12\%, n = 10 \text{ yr}) = 0.177$, the whole-module levelized cost is $(161\,000 \text{ USD} \times 0.177 + 4\,000 \text{ USD yr}^{-1}) / 90\,000 \text{ kWh yr}^{-1} \approx 0.36 \text{ USD kWh}^{-1}$ — three times the $0.12 \text{ USD kWh}^{-1}$ self-consumption tariff assumed in the revenue account (Section S5). The module therefore does not pay back through electricity alone: viability rests on the bundled power, crop, water, and carbon streams, which is precisely the multifunctional accounting the NEB formalizes. Second, discount-rate and tariff sensitivity: the NPV across $r = 8 - -15\%$ is tabulated in the SI (Table S6), and varying the electricity tariff over $0.10\text{--}0.20 \text{ USD kWh}^{-1}$ moves the simple payback from 4.64 yr to 3.39 yr, so the headline 4.32 yr is not an artifact of a single tariff choice. By contrast, the remaining barriers are physical rather than economic: fabrication of sub-2 nm NPoM gaps still requires laboratory-grade nanofabrication and a stable Au/graphene supply chain, and OPV encapsulation lifetime under tropical irradiance is the binding reliability constraint (Section S12.5). Subsidy policy alone cannot clear these barriers; they define the process-engineering roadmap that separates this modeled proof of concept from field deployment. Taken together, these sensitivities show that high-value crop selection and blended finance structurally transform the investment case.

2.10. Secure IoT telemetry via BB84 quantum key distribution

Field-to-cloud telemetry is secured using a simulated BB84 quantum key distribution protocol [30] over a buried single-mode fiber link, which eliminates the atmospheric turbulence and scintillation that affect free-space channels in agricultural environments. The telemetry layer is physically integrated with the NPoM-CQD sentinel network: the high-sensitivity biological and pesticide diagnostics drive high-stakes irrigation and crop-protection actuators, so hostile manipulation of those feedback commands could cause

artificial crop failure or pesticide contamination. Securing the commands with BB84 QKD therefore protects the integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [46]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a disclosed sample of at least 128 sifted bits, and the abort decision uses the 95 % Wilson upper confidence bound on that sample rather than the point estimate (at the nominal 4.8 % noise a 50-bit sample would give an upper bound of $\approx 13.5\%$, i.e. statistically indistinguishable from the threshold, whereas 128 bits give $\approx 9.9\%$). If that upper bound exceeds the Shor–Preskill unconditional-security threshold of 11 % [47], the protocol aborts the key exchange and halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [46], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration under secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

2.11. Experimental testability

The modeled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First, two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-modulator pulse shaping into dual-band channels at 750 nm and 820 nm can track the diagonal and cross-peak quantum-beat oscillations predicted here, verifying the extension of τ_c from 280 fs to 420 fs [48, 16]. Second, time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1} and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [28, 36]. More broadly, recent assessments identify quantum sensing for precision agriculture as an

539 emerging opportunity whose open calibration and robustness gaps match
540 the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295\text{ K}$, $L = 8$, $K = 2$. Rows are grouped by role: the upper block lists *input parameters fixed from* the cited experiments (calibration, not prediction), the lower block lists *out-of-sample observables* the model predicts (validation). Coherence lifetimes are reported for both excitations because the unfiltered calculation underestimates the room-temperature measurement by $\sim 30\%$ while the filtered calculation brackets it; quoting a single “passive” value would mix the two.

Parameter / Ob- servable		This work (calculated)	Experimental benchmark / lit- erature
<i>Parameters (calibration against literature)</i>			
Site coupling (cm^{-1})	J_{12}	-87.7	-87.7 ± 5.0 (2DES, Brixner 2005 [48])
Reorganization energy λ_D (cm^{-1})		35	35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50	50 (2DES, Panitchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2\text{ nm}^3$)	g_0	110 cm^{-1}	$120 \pm 15\text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [28]; g_0 scales from the mea- sured $g_{\text{ref}} = 120\text{ cm}^{-1}$ at $V = 1$ nm^3)
<i>Predictive observables (out-of-sample validation)</i>			
Coherence lifetime (broadband excitation)	τ_c	$280 \pm 25\text{ fs}$	$400 \pm 50\text{ fs}$ (2DES room temp., Pan- itchayangkoon 2010 [16])
Coherence lifetime (dual-band filtered)	τ_c	$420 \pm 35\text{ fs}$	$400 \pm 50\text{ fs}$ unfiltered benchmark (filter-enabled value to be measured by SLM-shaped 2DES)
SERS enhancement EF _{SERS} (relative, nor- malized at $V = 0.8$)		44 (at $V = 1.2$)	10^2 – 10^3 reported scaling for sub- nm NPoM; $\geq 10^6$ absolute single- molecule SERS [36]

541 In keeping with this grouping, the model is presently a *digital-twin-ready*
542 digital model: every parameter in the upper block is calibrated against pub-
543 lished measurements, only the lower-block observables are genuine predic-
544 tions, and no measured field telemetry has yet been assimilated — closing
545 that loop (assimilating in-situ 2DES/SERS and microclimate streams into
546 the parameter block) is the deployment step that would promote the model

547 to an operational digital twin.

548 3. Discussion

549 Our results demonstrate that quantum coherence engineering at the bio-
550 organic interface can deliver modeled sustainability benefits across micro-
551 scopic energy transfer, crop water use, carbon footprint, and data security.

552 3.1. Mechanism generality

553 The dual-band filtering effect is not specific to the FMO complex. The
554 resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property
555 of vibronically coupled chromophore aggregates [17]. We verified this by
556 applying the protocol to a minimal three-site excitonic trimer, obtaining
557 $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum
558 phase synchronization mechanism identified in photosynthetic antennas [27].
559 This suggests that any molecular aggregate satisfying the resonance condition
560 — including synthetic J-aggregates, light-harvesting complexes from plants,
561 and conjugated polymer domains in OPVs — could benefit from spectral
562 coherence protection. Vibration-steered energy flow of this kind has also
563 been demonstrated in a trapped-ion donor–acceptor simulator with tunable
564 dissipation [49].

565 3.2. The quantum divide

566 The advanced materials required for quantum agrivoltaics — graphene,
567 plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet ac-
568 cessible to subsistence farmers in developing economies. Our cooperative
569 cost-sharing model shows that, with a carbon-linked subsidy of 30 % and
570 a cooperative of 5 members, the payback period drops below 4yr, making
571 the technology economically viable under the modeled subsidy. We propose a
572 tiered subsidy framework in which the NEB itself serves as the verifiable met-
573 ric for subsidy qualification, linking quantum yield improvements to farmer
574 access.

575 3.3. Quantum digital twin

576 The modular architecture — quantum interface, FAO-56 microclimate,
577 LCA, IoT security — serves as the physics engine of an agrivoltaic *digital*
578 *twin*: a model-based supervisory replica that combines sensor telemetry from
579 the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend

580 irrigation schedules, OPV cleaning dates (from soiling factor [Eq. \(8\)](#)), and
581 carbon credit yields. The assimilation path from measurement back into the
582 model is concrete: each refresh cycle ingests (i) SERS intensity and Stern–
583 Volmer residual drift to update the calibration state and matrix-penalty fac-
584 tor, (ii) in-situ irradiance, vapor-pressure-deficit, and wind telemetry to re-
585 evaluate FAO-56 evapotranspiration and the soiling decay in real time, and
586 (iii) QBER statistics to gate the key schedule; periodic two-dimensional elec-
587 tronic spectroscopy would then refresh the coherence time and transfer yield
588 in the parameter block, closing the loop from field measurement back to the
589 Hamiltonian. This device-physics integration is distinct from the supply-
590 chain and breeding applications mapped in recent quantum-in-agriculture
591 reviews [\[11\]](#).

592 As a prospective extension, the scheduling layer of the digital twin
593 is formulated via the Quantum Approximate Optimization Algorithm
594 (QAOA) [\[50, 51\]](#), encoded as a 3-qubit variational quantum circuit of depth
595 $p = 3$ with 1024 measurement shots per macro-period ([Section S12.3](#)). The
596 algorithm encodes the irrigation–refrigeration–grid-export trilemma as a
597 quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and
598 solves it via a PennyLane emulator at sub-hourly timescales. Applied to
599 the daily solar profile of the Cameroonian scenario, the QAOA scheduler
600 yields water–energy allocations consistent with the 12–18 % gains over
601 classical greedy baselines reported in comparable implementations [\[51\]](#);
602 we note that with three binary decisions per macro-period the per-period
603 problem is exhaustively solvable, so the circuit is retained here as a validated
604 reference implementation (unit-tested against the brute-force optimum) and
605 as the migration path to the full 288-variable horizon, not as a claimed
606 quantum advantage at this scale. As a further prospective readout, our
607 quantum-gravimetry model converts the simulated water saving into a
608 groundwater mass anomaly: $\Delta g \approx 0.2 \mu\text{m s}^{-2}$ if the saved volume recharges
609 within the 500 m^2 footprint, falling to $\approx 3.2 \text{ nm s}^{-2}$ when spread over a 3 ha
610 recharge zone — the latter below single-instrument noise, which is why a
611 small regional gravimeter network or gradiometric common-mode rejection
612 is required ([Section S12.2](#)) [\[52, 53\]](#).

613 A hybrid quantum-classical signal processing pipeline — combining
614 Matrix Product State (MPS) denoising with quantum kernel ridge re-
615 gression (QKRR) [\[54, 55\]](#) — further enables the digital twin to extract
616 pre-symptomatic stress signatures from the noisy SERS and CQD fluo-
617 rescence telemetry. To satisfy edge computation constraints, the heavy

618 MPS tensor contraction and QAOA optimization run on the centralized
 619 cooperative server (or cloud), and the optimized weights are transmitted
 620 back to the field nodes — an edge–cloud split that mirrors the client–
 621 server decomposition of classical device-simulation workflows. The local
 622 ARM Cortex-M4 IoT node executes only the low-latency inference of the
 623 pre-trained, frozen-weight QKRR model. The MPS compressor acts as
 624 a nonlinear low-pass filter on the server that suppresses greenhouse noise
 625 (thermal fluctuations, biofouling, dust) while preserving the sharp vibronic
 626 features at 180 cm^{-1} and $1\,145\text{ cm}^{-1}$, which classical Savitzky–Golay filters
 627 tend to smear (Section S10). The cleaned spectra are projected through the
 628 quantum-inspired kernel for anomaly scoring, issuing early warnings that
 629 enable preemptive agronomic intervention before macroscopic symptoms
 630 appear — all within a 3.7 ms local compute budget on the IoT node. This
 631 millisecond-scale latency is not designed to match the multi-hour physical
 632 response of soil moisture percolation and plant transpiration; it enables
 633 spatial multiplexing instead, letting a single ARM Cortex-M4 edge processor
 634 sequentially query, compress, and classify SERS telemetry from dozens of
 635 distributed NPoM sensor nodes across the 500 m^2 greenhouse in under a
 636 second and reducing the electronic-component count. This chain connects
 637 microscopic quantum efficiency to field-level agronomic management; no
 638 existing agrivoltaic design integrates both [6, 2, 5].

639 3.4. *Prior art and positioning*

640 Previous agrivoltaic studies have focused on light-sharing through passive
 641 spectral filtering [6] or crop selection [2]. Semitransparent OPV greenhouse
 642 integration is already a modeled and experimental direction: semitranspar-
 643 ent organic solar cells can render a greenhouse energy-neutral in warm cli-
 644 mates while transmitting 32–45 % of the photosynthetically active band [56],
 645 stable semitransparent modules have been demonstrated on operating green-
 646 houses [7], and device records pair 14.6 % efficiency with 22 % average visible
 647 transmittance [57]. Our module assumption (15 % efficiency at 25 % aver-
 648 age visible transmittance) therefore sits at the upper edge of the reported
 649 efficiency–transmittance trade-off rather than beyond it. Our contribution is
 650 to add an active, quantum-coherent dimension to the spectral filter: the OPV
 651 is no longer a passive absorber but an active spectral shaper that prepares
 652 coherence-protected states in the photosynthetic apparatus. Multi-vector so-
 653 lar agri-systems combining agrivoltaics with hydrogen and desalination set
 654 the *Applied Energy* bar for water–food–energy integration [58]; we differ-

entiate through quantum-guided spectral co-design with integrated sensing, and we adopt the functional-unit discipline (emissions per kWh, economics per configuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this literature, the present work provides, to our knowledge, the first (i) agrivoltaic stack whose spectral architecture is co-designed with numerically exact open-quantum-system dynamics, trading excitation transport against the power-conversion budget along a single scanned design axis; (ii) monolithic PV-SERS device concept with field-relevant calibration in a single footprint; and (iii) multifunctional techno-economic and lifecycle framework that internalizes power, crop, water, sensing, and data-security trade-offs in $\$/W + \$/kg + \$/m^3$ terms. Table 4 condenses this positioning: no prior class delivers more than three of the seven functions in full (the sensing class delivers one), whereas the present stack delivers all seven in a single footprint under one accounting.

Table 4: **Capability matrix of agrivoltaic and agri-sensing approaches.** ✓ = function delivered within the stack; − = partial or incidental; — = absent. Classes: passive spectral claddings [6, 2, 7]; multi-vector agri-energy systems (agrivoltaics coupled to hydrogen and desalination) [58]; standalone agri-sensing hardware [10, 11, 12]. Multifunctional accounting denotes joint $\$/W + \$/kg + \$/m^3$ techno-economic and lifecycle evaluation of all functions. No prior class delivers more than three of the seven functions (partial entries are marked −); the present stack delivers all seven in a single footprint with unified accounting.

Function	Passive	Multi-vector	Sensing	This work
Electrical power (OPV)	✓	✓	—	✓
Crop microclimate	✓	✓	—	✓
Water management	−	✓	—	✓
Dynamic spectral control	—	—	—	✓
In-situ SERS sensing	—	—	✓	✓
QKD-secured telemetry	—	—	—	✓
Multifunctional accounting	—	—	—	✓

This approach is distinct from prior quantum biology studies [59] that treat the environment as a fixed noise source; here, the photonic environment itself is engineered through the NPoM cavity to maximize coherence protection. A key design choice warrants explicit justification: the NPoM cavity, despite imposing a $> 90\%$ penalty on excitonic transport (Table 2), is retained for one dedicated purpose: non-destructive, high-sensitivity SERS

675 readout of the FMO vibrational state, pesticide levels, and heavy-metal con-
676 tamination. The transport penalty is absorbed by confining NPoM operation
677 to the 1 % sentinel fraction (Equation (5)), preserving 97.1 % of the global
678 canopy trapping yield while maintaining full diagnostic capability.

679 3.5. Limitations

680 Several approximations constrain our analysis. The quantum dynamics
681 are restricted to the single-exciton manifold; under high solar flux, multi-
682 exciton states, singlet fission, or exciton-exciton annihilation in the OPV and
683 FMO interfaces could alter the collective coupling strength g_{eff} and transport
684 efficiency. The bath is treated as site-independent, and the reaction center
685 trapping is modeled as a phenomenological Lindblad-like rate. Two headline
686 observables remain predictions that no in-situ measurement has yet tested:
687 the filtered coherence lifetime ($\tau_c = 420$ fs, the +50 % gain) and the SERS en-
688 hancement factor ($\text{EF} \approx 44$ at $V = 1.2 \text{ nm}^3$). The benchmark table already
689 shows the *broadband* coherence time sitting ≈ 30 % below the unfiltered lab-
690 oratory value, which fixes the accuracy target for the first campaign: spatial-
691 light-modulator-shaped 2DES must first reproduce the unfiltered lifetime on
692 the actual device before the filtered value can be claimed as validated, and
693 a single-complex SERS measurement must confirm the enhancement factor
694 against the relative Purcell scaling we report. The FAO-56 model, while stan-
695 dard, does not capture transient stomatal responses to fluctuating light or
696 the dynamic microclimatic feedback between canopy transpiration and vapor
697 pressure deficit (VPD) inside the greenhouse envelope. Sensitivity analysis
698 over a plausible wind-speed range ($u = 1.5 - -2.5 \text{ m s}^{-1}$, i.e. ± 25 % around
699 the reference value) shifts the predicted reduction from 25.9 % to 30.4 % — a
700 ± 10 % relative band on the 28 % water saving, which propagates into a ± 10 %
701 range on the annual irrigation volume and corresponding NEB carbon offset.
702 The BB84 simulation assumes a buried single-mode fiber link; while buried
703 fibers are vulnerable to agricultural mechanical damage (plowing, root intru-
704 sion), a free-space optical (FSO) link would suffer from strong scattering un-
705 der dusty greenhouse environments, requiring dedicated atmospheric correc-
706 tion models. Software-based post-quantum cryptography offers a lower-cost
707 alternative, but the physical BB84 protocol provides information-theoretic
708 security immune to algorithmic cryptanalysis on resource-constrained em-
709 bedded nodes such as the ARM Cortex-M4 irrigation controllers. The 1 %
710 sentinel fraction is a design parameter; field optimization will require em-
711 pirical calibration of the SERS LODs under actual fouling conditions. We

conservatively estimate a $\times 10$ matrix-effect penalty on all reported LODs when transitioning from controlled laboratory buffers to raw agricultural sap or irrigation water, owing to competitive binding, background fluorescence, and ionic strength effects — i.e. field-basis limits of 500 nmol L^{-1} (oxidative stress), 10 nmol L^{-1} (2,4,5-T), 318 nmol L^{-1} (Pb^{2+}), and 8 nmol L^{-1} (NV-diamond pathogen readout); these field-basis values, not the buffer numbers, are the ones against which deployment thresholds must be set; the *Dynamic-Calibrator* compensates for this attenuation through real-time baseline correction. The zwitterionic polymer coating modeled on the NPoM graphene spacer (Section S12.1) acts as the primary fouling barrier in our simulations, with the *Dynamic-Calibrator* serving as a secondary correction layer; the dual architecture reduces the mandatory cleaning frequency, which is already factored into the cooperative OPEX model. We stress that the 60-day resilience of zwitterionic coatings applies strictly to sensors sheltered in protected micro-sampling chambers or rigid ceramic encasings; direct, unprotected exposure to abrasive soil matrices (bioturbation, quartz particles, tillage) would lead to mechanical scraping and rapid surface degradation.

The 30 % blended-finance subsidy required to achieve a 4.32yr payback period cannot be bridged by voluntary carbon credits alone (generating only 363 USD yr^{-1} , about 1.4 % of the net cash flow, at $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ for the 500 m^2 module); it must be underpinned by dedicated rural-adaptation finance channels such as the Green Climate Fund (GCF) and the African Development Bank’s Climate Change Fund (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance climate-resilient agriculture in Sub-Saharan Africa. Finally, the socioeconomic case is modeled for a single 500 m^2 floriculture archetype under one tariff, crop-price, and subsidy regime; the discount-rate, tariff, CAPEX, and joint Monte Carlo sensitivities reported above bound that choice for the modeled case, but transfer to other crops, sites, or cooperative sizes requires re-parameterization rather than direct extrapolation.

3.6. Outlook

Integrating quantum coherence engineering with agricultural sustainability opens a new axis for agrivoltaic design. Three experimental and theoretical directions are prioritized for future work.

First, on the hardware and materials scale, the quantum model must be extended to full multi-exciton dynamics under realistic solar spectra, accompanied by experimental validation via two-dimensional electronic spec-

749 troscopy (2DES) with spatial light modulator pulse shaping [60, 61]. Field tri-
750 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
751 sentinel network are required to evaluate in-situ sensor performance under
752 actual biofouling conditions and verify the coherence-preservation lifetime
753 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
754 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
755 ing Mie resonances [62] and nanocavity-free substrates — should be explored
756 to minimize the NPoM transport penalty while preserving SERS sensitivity.
757 Additional hardware extensions include passive thermal micro-shielding (Sec-
758 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
759 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
760 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

761 Second, scaling the digital twin to a multi-cooperative network will re-
762 quire hardware-in-the-loop QKD validation under dusty field conditions and
763 deployment of a real-time aquifer gravimetric array to confirm that the mod-
764 eled groundwater recharge connects at the satellite geodesy scale. Monte
765 Carlo uncertainty propagation in the full LCA — reported here with inde-
766 pendent input scatters — should next be extended with correlated climatic
767 input distributions drawn from regional reanalysis data.

768 Third, to prevent these advances from widening the “quantum divide,”
769 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
770 pillar certification framework covering QKD security, data sovereignty, sensor
771 detection limits, quantum fidelity, latency, and lifecycle sustainability (Sec-
772 tion S12.9). Our Cameroon cooperative proof-of-concept scenario serves as
773 the first application of this proposed framework, and returns an internal score
774 of six out of six pillars. We stress that this score is an internal consistency
775 check of the simulation against thresholds the authors themselves defined: no
776 third party has audited it, and it does not constitute a certification. Indepen-
777 dent experimental validation and independent audit will be required before
778 GQAS can be adopted as an operational standard. Scaling GQAS adop-
779 tion to smallholder networks across Sub-Saharan Africa and Southeast Asia
780 requires subsidized sensor supply chains, open-access certification auditing,
781 and regulatory recognition on premium export markets such as GlobalGAP.

782 4. Conclusions

783 We asked whether exact non-Markovian quantum dynamics can co-design
784 excitation transport with power conversion in a sensing-integrated spectral-

785 shaping agrivoltaic stack, with a positive multifunctional net benefit under
 786 field constraints. Within the modeled 500 m^2 cooperative case, the answer
 787 is affirmative but conditional: dual-band filtering raises the forward transfer
 788 yield by 25 % and extends coherence lifetime by 50 %; confining NPoM op-
 789 eration to a 1 % sentinel fraction preserves 97.1 % global canopy yield while
 790 retaining SERS diagnostics; the FAO-56 smart shield cuts evapotranspira-
 791 tion by 28 % with a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$; and the
 792 cooperative case pays back in 4.32 yr at 30 % subsidy. These claims are
 793 bounded by the $> 90 \%$ local NPoM transport penalty, the single-exciton and
 794 site-independent-bath approximations, the $\pm 10 \%$ microclimate uncertainty
 795 band, and the modeled (not field-validated) fouling and matrix-penalty as-
 796 sumptions. For deployment, the economic barrier is dominant and tractable:
 797 local-material construction and green-synthesis gold lower the capital cost
 798 from 322 USD m^{-2} to 140 USD m^{-2} (unsubsidized payback 2.68 yr). The
 799 physical barriers — nanofabrication of sub-2 nm NPoM gaps, in-field fouling
 800 of the sentinel fraction, OPV encapsulation lifetime under tropical irradiance
 801 — require process engineering rather than subsidies. Priority next steps are
 802 2DES validation of the filter protocol, low-loss dielectric antenna alternatives,
 803 field fouling trials of the sentinel network, and a localized Au/graphene sensor
 804 supply chain for Sub-Saharan manufacturing.

805 5. Methods

806 5.1. Proof-of-Concept scenario: 500 m^2 Floriculture Micro-Module

807 To ground the multi-domain framework in a concrete operational context,
 808 we consider a 500 m^2 high-tunnel greenhouse located in Cameroon’s West-
 809 ern Highlands. This region supports high-value floriculture (strawberries, cut
 810 flowers) suitable for micro-module deployment by cooperatives. The environ-
 811 ment features a mean annual solar irradiance of $5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$ [63] and
 812 significant biofouling potential that necessitates the NPoM *DynamicCalibra-*
 813 *tor* and regular OPV cleaning (factored into the cooperative OPEX model).
 814 Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotran-
 815 spiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the
 816 modeled water-savings credit of $460 \text{ L m}^{-2} \text{ yr}^{-1}$ ($230 \text{ m}^3 \text{ yr}^{-1}$ for the 500 m^2
 817 greenhouse; Section S5). The area-weighted device-level excitation-transfer
 818 yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) stays near the passive baseline while accommodating a
 819 1 % NPoM sentinel fraction for SERS diagnostics; it is a quantum-transport
 820 metric for the device, not a crop-yield proxy, because harvestable biomass

821 is set by the AVT/PAR transmission budget of the shield (Section 5.1) and
 822 no coherence-derived gain enters the crop account. Crop production is as-
 823 sumed at the horticultural standard for the crop mix ($8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$); the
 824 quantum-fertilizer channel adds $0.96 \text{ kg m}^{-2} \text{ yr}^{-1}$ (8 % biostimulation of the
 825 $12 \text{ kg m}^{-2} \text{ yr}^{-1}$ reference biomass that sets the Scenario A biomass bar in
 826 Figure 4b, not of the $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ monetized floriculture yield; Scenario
 827 A only), and no coherence-derived yield gain is monetized — the passive-
 828 fraction coherence gains reported in the companion study [25] are treated
 829 as a device-physics result, not a crop-revenue stream. The separation is
 830 deliberate and worth stating plainly: the quantum layer enters the NEB
 831 and cash-flow accounts on the cost side only (the NPoM sentinel and QKD
 832 modules are priced as capital and operating line items) and as preserved
 833 function (0.971 global trapping yield with full diagnostics on the 1 % sen-
 834 tinel), while the benefit side is carried by grid displacement, water saving,
 835 and crop revenue. The avoided crop loss that pre-symptomatic SERS and
 836 NV-diamond detection could deliver is deliberately *not* monetized, so the
 837 reported NEB and payback are conservative with respect to the diagnos-
 838 tic function. The OPV module generates 90 MWh yr^{-1} of electrical energy
 839 for a 500 m^2 installation at $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$ — the guaranteed-yield ac-
 840 counting ($5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$ irradiance, 15 % module efficiency, 0.57 perfor-
 841 mance ratio) that is used consistently in both the revenue and the carbon
 842 accounts. Two accounting choices are stated explicitly here. First, the
 843 two transmitted passbands (750–820 nm, 100 cm^{-1} each) together carry only
 844 10.2 W m^{-2} of the ASTM G173-03 global-tilt reference spectrum — 1.0 %
 845 of AM1.5G and 1.3 % of the 280–1100 nm harvestable window (computed
 846 from the ASTM G173-03 global-tilt spectrum) — and are priced into the
 847 module’s 25 % average visible transmittance rather than deducted a second
 848 time; had they been carved out of an otherwise opaque module, the penalty
 849 would be $\leq 1.3\%$ on generation ($90 \text{ MWh yr}^{-1} \rightarrow \sim 88.9 \text{ MWh yr}^{-1}$), i.e.
 850 NEB $72.6 \rightarrow \sim 71.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and payback $4.32 \text{ yr} \rightarrow 4.35 \text{ yr}$, which
 851 we take as the bounded worst case of the spectral co-design. Second, displac-
 852 ing grid electricity at $450 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ — slightly below the $480 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$
 853 global average for 2023 [64], so our headline factor is conservative relative
 854 to the world average but high for a hydro-dominated national grid — offsets
 855 $40.5 \text{ t}_{\text{CO}_2\text{e}} \text{ yr}^{-1}$ ($81 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$); the NEB of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ is this
 856 displacement plus the $0.14 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ water-pumping credit, minus the
 857 lifecycle footprint of the stack ($8.5 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ for the quantum configu-
 858 ration and $5.0 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ for the static-shading configuration). The dis-

859 placement alone therefore exceeds the lifecycle footprint, so the system is net-
 860 negative on operational carbon, and the NEB remains positive for any grid
 861 factor above $\approx 47 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$: a hydro-heavy grid at $200 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ would
 862 still yield $\sim 27.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. For the deployment country the relevant
 863 factor is Cameroon’s national power-sector intensity of $\approx 290 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$
 864 (2019, 62 % hydropower, projected to reach 75 % by 2023) [65], which yields
 865 $\approx 43.8 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ — still six times the $\approx 47 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ break-even fac-
 866 tor, so the net-benefit conclusion is robust to using the national rather
 867 than the generic grid factor. At $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ [66], the system generates
 868 363 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched
 869 to cooperative ownership models: under a 30 % blended-finance subsidy, the
 870 payback shortens from 6.17 yr unsubsidized to 4.32 yr, with each of the five
 871 cooperative members exposed to only 22 540 USD of net capital. Under these
 872 modeled assumptions, quantum-enhanced agrivoltaics can be economically
 873 viable when coupled with strategic crop selection and robust field-sensor in-
 874 tegration.

875 5.2. *PT-HOPS/SBD quantum dynamics*

876 All quantum dynamics simulations use the Process Tensor Hierarchy of
 877 Pure States with Stochastically Bundled Dissipators as implemented in the
 878 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
 879 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2 \text{ fs}$, established as the
 880 production standard for converged dynamics at 295 K [25]. Static disorder
 881 is modeled as Gaussian fluctuations of the site energies with standard devi-
 882 ation $\sigma = 50 \text{ cm}^{-1}$, sampled over $n = 100$ realizations. The bath spectral
 883 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
 884 Drude–Lorentz background ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) and 12 under-
 885 damped vibronic modes (frequencies 180–1 500 cm^{-1} , Huang–Rhys factors
 886 $S_k = 0.003 - -0.050$).

887 5.3. *FAO-56 evapotranspiration*

888 Crop evapotranspiration is computed using the standard FAO-56
 889 Penman–Monteith equation [29] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408 \Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

890 with R_n attenuated by the array-level shading factor $f_{\text{shade}} = \text{GCR}(1 -$
 891 $\tau_{\text{AVT}}) = 0.41$ (ground-coverage ratio $\text{GCR} = 0.55$, average visible trans-
 892 mittance $\tau_{\text{AVT}} = 25\%$), evaluated on daily net radiation $R_n = R_{ns} - R_{nl}$
 893 with $R_{ns} = 0.77R_s$ and $R_{nl} \approx 1.9 \text{ MJ m}^{-2} \text{ d}^{-1}$ for the reference day ($R_s =$
 894 $20.74 \text{ MJ m}^{-2} \text{ d}^{-1}$, i.e. $5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$). The aerodynamic term uses the
 895 standard FAO-56 wind expression with no empirical rescaling. We note that
 896 the modified FAO-56 formulation represents a first-order approximation; un-
 897 der the enclosed high-tunnel envelope, boundary layer resistance and sensible
 898 heat exchange are altered relative to open field conditions. Future iterations
 899 of the digital twin should integrate three-dimensional Computational Fluid
 900 Dynamics (CFD) modeling to map localized microclimatic gradients and la-
 901 tent heat transfer. Wind speed is held static at the free-stream reference value
 902 $u = 2.0 \text{ m s}^{-1}$ and is evaluated identically for the open-field and the shielded
 903 field, so that the reported 28 % reduction isolates the radiation effect of the
 904 array rather than a ventilation difference; the bracket $u \in 1.5 - 2.5 \text{ m s}^{-1}$
 905 carries the $\pm 10\%$ sensitivity band quoted in the Limitations. Temperature
 906 is fixed at 25°C and relative humidity at 60 % unless specified otherwise.

907 5.4. BB84 QKD simulation and fail-safe IoT architecture

908 The BB84 protocol is implemented as a Python simulation of polarization-
 909 state preparation, noisy transmission over a buried single-mode fiber channel,
 910 sifting, and QBER estimation. Alice prepares $4 \times N$ random bits (for $N =$
 911 256 key length) in random rectilinear or diagonal bases. Bob measures in
 912 random bases, and matching-basis events are retained for key sifting. QBER
 913 is estimated on a disclosed sample of at least 128 bits (25 % of the 512 sifted
 914 bits), using the 95 % Wilson upper confidence bound as the decision statistic.
 915 If that upper bound exceeds the Shor–Preskill threshold of 11 % [47], key
 916 generation aborts, halting the QKD-protected irrigation command relay. A
 917 last-validated soil-moisture reading from the *DynamicCalibrator* is used to
 918 execute a conservative local irrigation fallback.

919 5.5. SERS enhancement scaling limits

920 The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode vol-
 921 ume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of
 922 the localized electromagnetic field. We note that this standard $|E|^4$ approx-
 923 imation represents a lower-bound semi-classical estimate. In sub-nanometer
 924 picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling,

925 and non-local dielectric screening alter the field distribution relative to clas-
 926 sical electrodynamics, while molecular optomechanical coupling can pump
 927 vibrational modes out of thermal equilibrium. Precise calculation of the
 928 localized fields in this regime requires non-local Green’s tensor integration,
 929 which motivates the proposed in-situ experimental validation. The bench-
 930 mark against the experimentally reported 10^2 – 10^3 range for sub-nanometer
 931 NPoM gaps [36] is given in Table 3.

932 5.6. *DynamicCalibrator, thermal shielding, and soiling factor*

933 Long-term sensor baseline drift due to biofouling and thermal cycling
 934 is corrected at three hardware and algorithmic levels. The primary fouling
 935 barrier is the zwitterionic polymer functionalization layer deposited on the
 936 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
 937 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
 938 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
 939 micro-shielding layer ($1\text{ }\mu\text{m}$ to $5\text{ }\mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the
 940 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
 941 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the
 942 probe experiences only 15% of the ambient greenhouse temperature fluctu-
 943 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25%
 944 to 0.79% under a $\pm 15\text{ K}$ diurnal cycle. The *DynamicCalibrator* acts as a
 945 tertiary algorithmic correction: an exponential moving average (EMA) with
 946 smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor
 947 $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the
 948 EMA $\hat{\mu}(t)$ exceeds 20% of the initial baseline, triggering OPV panel cleaning.
 949 At the panel level, OPV electrical output is attenuated according to
 950 Eq. (8), with a default decay rate of $0.5\text{ }\text{d}^{-1}$ and a minimum soiling factor
 951 of 0.75 (enforced floor at 25% maximum efficiency loss).

952 *Reproducibility and computational cost*

953 All stochastic streams are seeded: the PT-HOPS/SBD propagator draws
 954 from a fixed MesoHOPS random seed (42) for the static-disorder and noise
 955 realizations, and the joint uncertainty analysis reported above uses a fixed
 956 Monte Carlo stream (seed 42), so every quoted number is regenerable bit-for-
 957 bit from the released configuration. A single production trajectory (hierarchy
 958 depth $L = 8$, $K = 2$, $\Delta t = 0.2\text{ fs}$, 1 ps window) requires on the order of
 959 45 min on one modern 3 GHz core; the archived campaign behind the tables
 960 of this work comprises several hundred such trajectories, i.e. an order of

961 10^3 CPU-hours on the 48 core-class workstation acknowledged below. The
 962 $L = 8$ quantum engine is executed offline: the digital twin's 60 s refresh
 963 cycle reads the archived and surrogate observables (population trajectories,
 964 yields, calibration states) rather than re-propagating the hierarchy in real
 965 time, which is why the deployed refresh rate is compatible with operational
 966 irrigation control. Trajectory counts are heterogeneous by design and are
 967 stated with every quoted value: $n = 100$ at the reference volume $V = 0.8$
 968 nm^3 , $n = 20$ for the $V = 1.2 \text{ nm}^3$ production run and for the independent
 969 re-simulation of all seven mode volumes behind [Figure 1](#), and $n = 2$ for the
 970 remaining scan points and for the 77 K case (indicative only); no ensemble
 971 statistic in this work is quoted without its n , and the differing counts are
 972 carried as an explicit caveat wherever configurations are compared.

973 *Data availability*

974 All simulation parameters, YAML configuration files, orchestration
 975 scripts, and analysis code needed to reproduce every figure and table are
 976 available from the corresponding author upon reasonable request and will
 977 be deposited in a public repository with a DOI upon acceptance. The
 978 PT-HOPS/SBD implementation is based on the open-source MesoHOPS
 979 library (v1.7) [\[20\]](#).

980 *Code availability*

981 The digital-twin simulation framework (orchestrator, QAOA scheduler,
 982 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
 983 pathogen sensor, and GQAS compliance checker) and all simulation param-
 984 eters, including the 12-mode vibronic bath spectral density and the 8-site
 985 FMO electronic Hamiltonian specified in the machine-readable parameter
 986 configuration file (v1.4), are available from the corresponding author upon
 987 reasonable request and will be deposited in the same public archive with a
 988 DOI upon acceptance.

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994 **CRedit authorship contribution statement**

995 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
996 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
997 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
998 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
999 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
1000 editing.

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1002 The authors declare that they have no known competing financial inter-
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1009 **Declaration of generative AI and AI-assisted technologies in the**
1010 **writing process**

1011 During the preparation of this work the authors used the following
1012 AI-assisted tools: OpenAI ChatGPT (GPT family large language model)
1013 and Anthropic Claude (Claude family large language model), applied solely
1014 to language editing, LaTeX formatting support, and reference-formatting
1015 checks. No scientific content, data, analysis, or interpretation was generated
1016 by these tools. All scientific content was reviewed and verified by the
1017 authors, who take full responsibility for the content of the publication.

1018 **Nomenclature**

	Symbol	Meaning
	Φ_{FT}	Forward (reaction-center) transfer yield
	τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
	λ_{res}	Residual reorganization energy of polaron-dressed states
	g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
	κ / Q	Plasmon linewidth / quality factor
1019	$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
	Γ_{RC}	Reaction-center trapping rate
	$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
	$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
	NEB	Net Ecological Benefit
	$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
	QBER	Quantum bit error rate (BB84 link)
	Δg	Groundwater gravity anomaly
	$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

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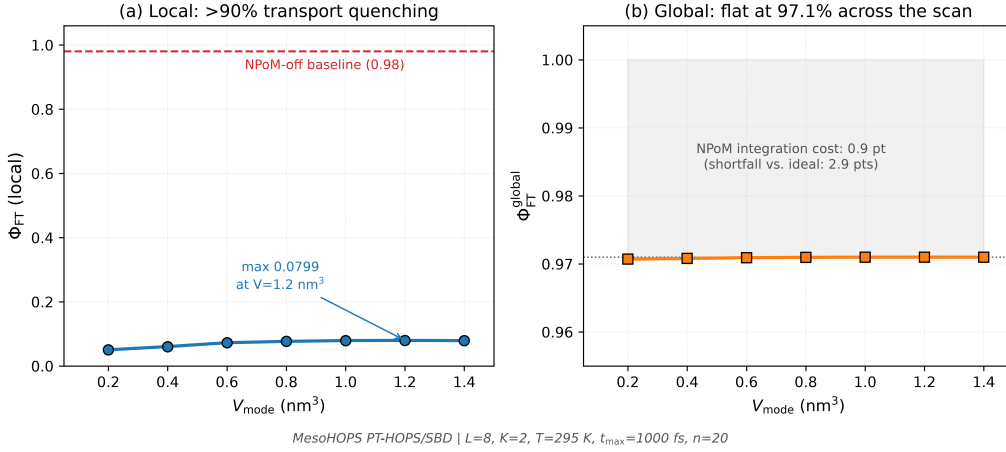


Figure 1: **Local quenching, global stability: the central result in one picture.** (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295 \text{ K}$): locally, transport is quenched by more than 90 % relative to the pristine FMO baseline (0.98), with a factor-1.6 variation across the accessible range and a maximum of 0.0799 at $V_{\text{mode}} = 1.2 \text{ nm}^3$ (Table 2). (b) Area-weighted global canopy trapping yield for the 1 % sentinel fraction (Equation (5)): flat at 0.971 across the entire scan — the deep local suppression is absorbed at a constant 0.9-point canopy-scale cost (0.980 to 0.971), leaving the canopy 2.9 points below an ideal yield of 1.0. The flat curve in (b), not the quenching in (a), is the design-relevant outcome: diagnostics integrate into the stack without compromising productivity. Provenance: the curves are plotted from the independent $n = 20$ re-simulation of all seven mode volumes; the tabulated values in Table 2 are the original production runs ($n = 2$ except where noted) and the re-simulation reproduces them within ± 0.002 .

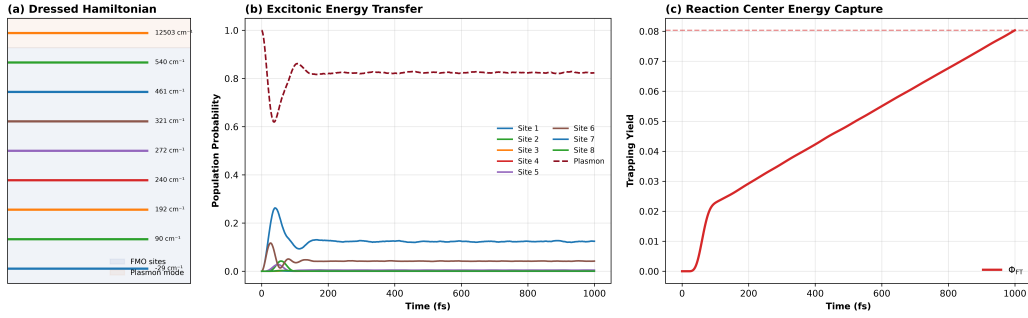


Figure 2: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Population dynamics of the $n = 20$ production run at $V_{\text{mode}} = 1.2 \text{ nm}^3$ and 295 K (the scan optimum of Table 2): the plasmon mode (Site 9) loads within the first 10 fs and captures 83 % of the excitation, starving the trapping sites — the microscopic origin of the 91.8 % local penalty. (c) Cumulative reaction-center trapping yield $\Phi_{\text{FT}}(t)$ for the same run, saturating at 0.0799. All data: $L = 8$, $K = 2$, $n = 20$ disorder realizations (Section S11). The passive-fraction dynamics underlying the coherence and yield gains of Figure 2c are reported in [25]. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield.

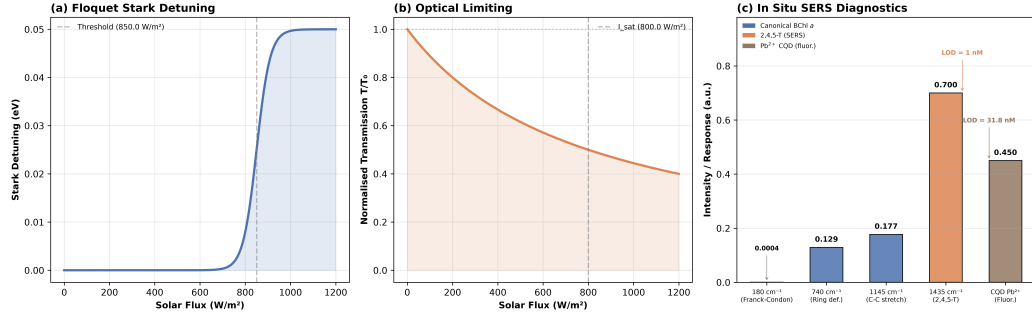


Figure 3: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux: the hysteretic switch closes the resonant site-1 and site-6 transfer channels when the flux rises above 850 W m^{-2} and reopens only below 750 W m^{-2} ($\Delta = 403 \text{ cm}^{-1} \gg |J_{12}| = 87.7 \text{ cm}^{-1}$, single-channel transmission ~ 0.05) — the solid-state analogue of non-photochemical quenching. (b) Flux-dependent optical limiting of the 750 nm and 820 nm passbands: at 1000 W m^{-2} the two filter passbands themselves close to 44 % of baseline, protecting the exciton budget precisely where the co-designed filter concentrates it. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A Dynamic-Calibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

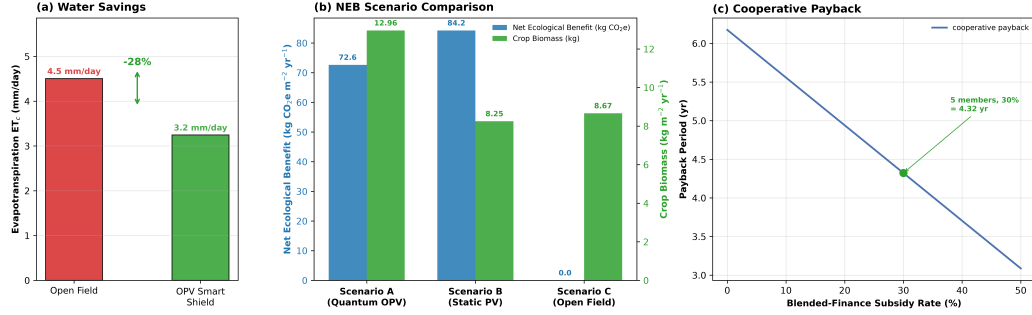


Figure 4: **Water–Energy–Food network assessment.** All panels refer to the 500 m^2 high-tunnel floriculture micro-module in Cameroon’s Western Highlands described in [Section 5.1](#). **(a)** FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. **(b)** Net Ecological Benefit comparison across three operating scenarios: Quantum OPV (A: OPV–NPoM shield with dual-band transmission and the quantum-fertilizer biomass boost), Static shading (B: opaque static-shading module — higher PV output, suppressed canopy), and Open field (C: no module). Scenario A achieves $72.6 \text{ kg}_{CO_2e} \text{ m}^{-2} \text{ yr}^{-1}$ with an effective crop biomass of $13.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $84.2 \text{ kg}_{CO_2e} \text{ m}^{-2} \text{ yr}^{-1}$ but only $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass (plotted bar values 12.96, 8.25 and 8.67, rounded in the text). The comparison is deliberately uncomfortable: an opaque static module still buys more carbon than the quantum shield, but it does so by suppressing the canopy to 84 % of reference biomass, whereas the quantum shield preserves near-baseline productivity (0.971 global trapping yield, 13.0 including biostimulation) — exactly the food–energy trade-off that motivates spectral co-design. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. **(c)** Simple, undiscounted payback period of the module as a function of the blended-finance subsidy rate (five-member cooperative; the discounted NPV and its discount-rate sensitivity are reported in the text and [Section S5](#)). At the modeled 30 % subsidy, payback = 4.32 yr (net annual cashflow of $26\,075 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modeled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

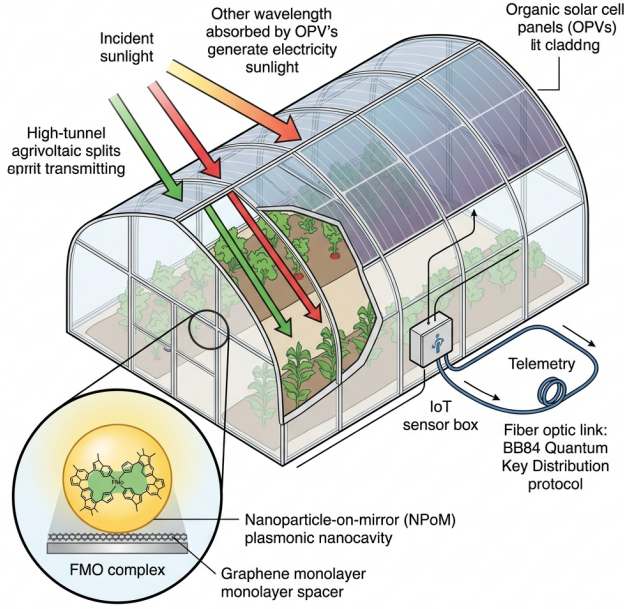


Figure 5: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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